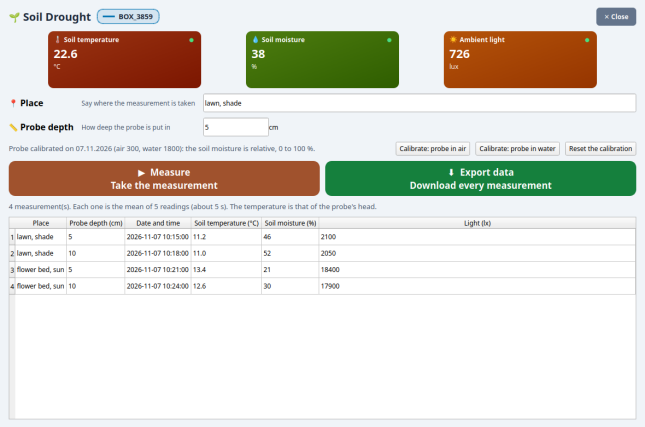
Soil Drought

How dry is the ground around the school? A survey by place and by depth

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** one 50-minute lesson, mostly outdoors; the analysis in Excel in a second lesson or at home
• **Sensors:** the STEMMA soil probe; the light sensor (VEML7700 or BH1750), optional



The measuring page after a few measurements. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). In year V (geography: soil) and year VI (biology: plants and water) pupils measure, fill a table and compare places; at upper secondary level (geography, year IX: water in the soil, climate risks) they add the averaging of repeats, the spread between them and a chart made in Excel.

Assumed knowledge

Reading a table, the idea of a percentage and of an average. The lesson *Soil and Water* helps but is not required: this screen calibrates the probe too.

Time in class

One 50-minute lesson: calibration and prediction in the classroom (10), the survey outdoors (30), back inside to check the file (10). The chart and the discussion fit into a second lesson or a homework.

Teacher preparation

15 min — choose three or four places in the school grounds that differ: a lawn in the sun, ground under a tree, a trodden path edge, a flower bed. A small trowel to open a hole for the deeper reading, a ruler, a glass of water for the calibration, paper towels or a cloth, and a laptop with a charged battery. Note how many days have passed since the last rain.

Sensor settling time

The probe gives one value a second. After pushing it in, wait until the value on the screen stops moving (a few seconds), then press **Measure**: the app averages the next five readings, about five seconds, during which nobody touches the probe.

Sensors and boxes The **STEMMA soil probe** (Adafruit), read by firmware 2.5.0 or newer: a raw capacitance value, about 200 in air and up to about 2000 in water, different from one probe to the next. Its chip also reports a temperature, shown as *Soil temperature*; it is the temperature of the probe's head, good to about ±2 °C at best. The **light sensor** (VEML7700, or BH1750 when the box has that one) records how bright it was at the moment of each measurement. Classroom boxes without a probe cannot run the activity.

Three routes

Short	Two places, one depth, three repeats each: 20 minutes outdoors, enough to compare sun and shade.
Full	Three or four places, two depths, three repeats: 24 to 32 rows in one lesson, then the chart in Excel.
Project	The same places once a week, or the day after rain and again a week later, with the same box and the same calibration: how fast does each place dry out?

The application starts in English. Buttons are named here exactly as they appear on the screen.

The screen is hidden: switch it on once in Settings

The activity needs the soil probe, which the usual boxes do not have, so the app does not show it by default. In **Settings**, under **Visible activities**, tick *Soil Drought*; it appears at once on the **Activities** page, among the **Explore tools**. The tick stays on that computer.

1. Where this fits in the curriculum

- **Geography** — in year V, soil and its layers; in year IX, water in the soil and drought as a climate risk: why the same dry week hurts a field in one place and hardly shows in another.
- **Biology** — in years VI–VII, plants and water: roots take water from a particular layer, and a lawn dries out from the top down; in year X, soil as a living environment.
- **Mathematics** — the average of three repeats, the range as a first measure of uncertainty, a bar chart grouped by place and depth.

The core skill is **a fair survey**: one probe, one calibration, the same way of inserting it everywhere, and enough repeats to tell a real difference from the scatter between neighbouring spots.

2. Time plan

Stage	Time	What happens
Calibration and prediction	10 min	In the classroom: the probe in air, then in water. Pupils mark the places on a sketch and predict which is driest.
The survey	30 min	At each place, two depths, three spots a hand's width apart: one **Measure** each.
Back inside	10 min	The table on screen, the file in `exports`, **Export data** to Excel.
Analysis (2nd lesson)	30–50 min	Averages, a bar chart by place and depth, the discussion of drought.

3. Materials

Item	Per group	Notes
MagicBox with the soil probe	1	The light sensor on the same box, if you have one.
Laptop with the app, charged	1	The screen has to be ticked once in **Settings** → **Visible activities**.
Small trowel or garden dibber	1	To open a hole for the deeper reading.
Ruler	1	For the depth of the probe's tip below the surface.
Glass of water	1	For the calibration, in the classroom.
Paper towels or a cloth	1	To wipe the probe between spots.
Sketch of the school grounds, pencil	1	The places, marked A, B, C, D.

4. Scientific background for the teacher

What the probe measures, and why drought starts at the top

The probe measures the **electrical capacitance** between two electrodes on its board, which depends on the dielectric permittivity of what surrounds them. Water has a permittivity of about 80, the minerals of the soil about 4–8, air 1: the more water around the probe, the higher the raw number. That number also depends on how well the soil touches the board, on salt and fertiliser, and on the probe itself.

A relative scale. Pupils capture the value in air (a) and in water (w); for each reading x the app computes

$$u = \frac{x - a}{w - a} \cdot 100\,\%,$$

limited to between 0 and 100 %. This is **not** the water content of the soil as a soil scientist reports it, in cubic centimetres of water per cubic centimetre of soil; it is a position between two fixed points. Wet soil seldom comes near 100 %, because even saturated soil is mostly mineral. The scale is good for comparing places and depths measured with the same probe and the same calibration, and that is how the lesson uses it.

How soil dries. After rain, water fills the pores and the surplus drains down. Evaporation then takes water from the top few centimetres, and roots take it from the layer where they grow. In a dry spell the surface layer dries first, so the top is often drier than the soil 10–15 cm down; just after rain the order can be the reverse. Sun and wind speed up evaporation; shade slows it, but a tree's roots and crown, which catches part of the rain, can leave the ground under it drier than an open lawn. Compacted soil has fewer and smaller pores: rain runs off instead of soaking in.

Drought. A dry spell with little rain is a meteorological drought; when the soil no longer holds the water plants need, scientists speak of a soil-moisture (agricultural) drought. The second follows the first with a delay, and how long the delay is depends on the soil and on what grows on it, which is what the pupils see in their own grounds.

4.1 What to expect

Place	After a dry week	Why
Lawn in the sun	top drier than below	evaporation and grass roots in the top layer
Under a tree	often dry at both depths	the crown caught rain; the roots draw water
Trodden path edge	dry, hard to push into	little rain soaked in; stop if the probe does not go in
Watered flower bed	the wettest	watering and loose soil

These are tendencies, not rules: the last rain, the type of soil and how the grounds are kept change them. A difference counts only if it is clearly larger than the spread of the three repeats at one spot.

5. Lesson flow

The screen has two pages: a start page that lists the sensors and a measuring page with the place, the depth and the **Measure** button.

1. **The question (3 min).** How many days since it last rained? Where would a plant suffer first? Pupils mark three or four places on the sketch.
2. **Start and calibration (7 min).** On the start page, the *Soil probe* and *Light (VEML7700 / BH1750)* tiles turn green with *The box sees it*; then **Start**. On the measuring page: the probe dry, held in the air touching nothing, **Calibrate: probe in air**; then dipped in water up to its line, **Calibrate: probe in water**.

Technical note — the calibration, a measurement and the file

The calibration. Each button takes the mean of the last five seconds of readings. The water value must be at least 200 above the air value, otherwise the app refuses it and asks again. The result is kept for each box, on that computer, and shared with the lesson *Soil and Water*; the line above the buttons reads *Probe calibrated on 07.11.2026 (air …, water …).* **Reset the calibration** deletes it after an *Are you sure?*. Without a calibration the card shows *Soil probe (raw)* with the raw number, and the `Soil moisture (%)` column stays empty.
One measurement. **Measure** turns into *Measuring…*; the app waits for the next five readings, one a second, and writes one row: the means of moisture, temperature and light, with the place and the depth typed in the two fields.
The file. Every row is saved at once to `MagicBox\exports\soil_drought_<YYYY-MM-DD_HHMM>.csv`, so closing the window loses nothing. **Export data** saves a copy as an Excel workbook (sheets `Data` and `Legend`) or as a CSV. The column names are English in every language: `Place`, `Probe depth (cm)`, `Timestamp`, `Soil temperature (°C)`, `Soil moisture (%)`, `Light (lx)`; the table on screen shows the same columns in the app's language. A CSV saved with the app in English uses commas and a decimal point; in Romanian, French or German, semicolons and a decimal comma.
The raw values. The box's own log, opened from **Data & Export**, keeps the raw `SOIL` value of every second; a session measured without a calibration can be recovered from it.

3. **The prediction (2 min, before going out).** *Which place is driest? Is the top drier than the layer below?*
4. **The survey (30 min).** At each place: type the place, always spelt the same way (*A lawn sun*); push the probe in up to its line, measure the depth of its tip with the ruler and type it in **Probe depth**; wait for the value to settle; **Measure**. Three spots a hand's width apart,

wiping the probe each time. Then open a small hole about 10 cm deep with the trowel, push the probe into its floor and repeat the three measurements with the new depth. Leave the box on the ground beside the probe, the light sensor facing up and nobody's shadow on it.

5. **Back inside (10 min).** Check the table on screen: one row per click, the places spelt alike, no empty moisture column. **Export data** to an Excel workbook for the analysis.

The investigation, in one table

The question. Where is the soil around our school driest, and is the top layer drier than the one below it?
The hypothesis. The order of the places and of the two depths, written on the sketch before going out.
What I change. The place and the depth of the probe.
What I measure. The relative soil moisture; the light at the moment of the measurement as a sign of sun or shade.
What I keep the same. The probe and its calibration, how it is pushed in, the time of day (the whole survey within one lesson), the days since rain.
How many repeats. Three spots at each place and depth. The average is the result; the range (largest minus smallest) shows how much one spot differs from its neighbour.
What makes a reading valid — and what makes it an artefact. The board in full contact with the soil, no stone against it, no air gap left by wiggling it in. A probe still wet from the last hole, a puddle, or soil with fertiliser gives a value that is too high; a gap gives one that is too low.

6. Analysis after the activity

6.1 Questions to ask afterwards

1. Which place was driest at the surface? And at 10 cm? Is it the same place?
2. At which place were the three repeats furthest apart? What could make two spots a hand's width apart so different?
3. Was the light column high where you expected sun? Did a cloud pass during the survey, and how would you notice it in the data?
4. If it does not rain for another two weeks, which plants in the grounds will suffer first, and why?

6.2 Assessment ideas

- **Low stakes:** the pupil fills the table, works out the averages and puts the places in order.
- **Standard:** the pupil draws the bar chart in Excel and explains one difference between places or depths with evaporation, roots or compaction.
- **Extension:** the pupil says which differences are larger than the range of the repeats and which are not, and explains why the percentage is relative to the calibration.

Common pitfalls and misconceptions

Depths typed differently. *10* is saved as a number, *10 cm* as text, and Excel puts them in different groups. Type only the number, and the same number for the same depth.
The place spelt in several ways. *Lawn sun* and *lawn in sun* are two places for Excel. Agree on the names before going out.
Forcing the probe. The board breaks before hard ground gives way. Loosen a hole with

the trowel, or note *too hard* on the sketch; that is a finding too.
Reading the temperature as the soil's. The chip sits in the probe's head, near the surface and in the sun; its value follows the air and the sunshine more than the soil.
A calibration done with a wet probe. The air value comes out too high, and every percentage too low. Wipe the probe first.

Safety and classroom management

Weather and electronics. No survey in the rain; the box and the laptop stay dry, on a bag or a jacket rather than on wet grass.
Hands. Wash hands after handling soil; gloves where there may be litter or animal droppings.
The grounds. Stay away from car parks and driveways; the trowel is carried point down. Fill in the holes afterwards.

Adaptations for pupils with special educational needs

One job each. One pupil types, one inserts and wipes the probe, one holds the ruler, one reads the value aloud and ticks the sheet.
No calculation. Younger pupils compare only the order of the places, from driest to wettest, on the sketch.
Mobility. Choose places along a paved path; the probe on its cable reaches the soil while the pupil stays on the path.

If things do not go to plan

No time to calibrate. Measure anyway: the card shows the raw value. Pupils write it on the sheet, and the raw `SOIL` column in the box's log from **Data & Export** keeps it too. Compare places with the raw numbers; the same probe makes them comparable.
No light sensor. The light column stays empty; write *sun* or *shade* into the place name.
The laptop cannot go outside. Dig up a small sample from each place and depth into labelled jars with lids, and measure them in the classroom the same day.
The box will not connect. Use the *What to expect* table and the pupils' predictions as the data set, and plan the survey for another day.

7. Further resources

- **European Drought Observatory (JRC)** — `https://edo.jrc.ec.europa.eu` — maps of soil moisture anomaly across Europe, updated every ten days; compare your grounds with the map of your region.
- **GLOBE Program** — `https://www.globe.gov` — search for *soil moisture protocol*: how schools around the world measure soil water.
- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — recent rainfall, to count the days since the last rain.
- **Adafruit** — `https://learn.adafruit.com` — search for *STEMMA soil sensor*, for how the probe works.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *if it does not rain for two more weeks, where in our grounds will a plant suffer first, and what in your data tells you so?*

On the **common three-axis rubric**:

- **Measurement practice** — the pupil calibrates the probe, inserts it the same way everywhere and records three repeats.
- **Data interpretation** — the pupil averages the repeats, draws the chart and puts the places and depths in order.
- **Scientific reasoning** — the pupil explains a difference with evaporation, roots or compaction and states a limitation: the percentage is relative, and one afternoon is one moment.

The rubric itself, with descriptors for all levels, is defined in the *curriculum plan*.

9. Pupil sheet

The sheet is a separate document, in `docs/en/student_sheets`: the file `soil_drought.pdf`. It has the usual layers: what I see on the screen, what I do on my own, the conclusion I write. The table has one row for each place and depth, three columns for the repeats and one for the average.